Recitation: Function Operations Review

Problem 1. Suppose $f(x)$ and $g(x)$ are the functions given below.

$$f(x) = 2x - 3$$

$$g(x) = x^2 + 2x - 1$$

Compute the following:

(a) $f(5) =$

(b) $g(-2) =$

(c) $f(0) - 3g(2) =$

(d) $(f - g)(x) =$

(e) $(fg)(x) =$

(f) $\left(\dfrac{f}{g}\right)(x) =$

(g) $(f \circ g)(0) =$

(h) $(g \circ f)(0) =$

(i) $(f \circ g)(x) =$

(j) $(g \circ f)(x) =$

Problem 2. Find the average rate of change for $f(x) = \dfrac{4-5x}{3}$ for $-1 \le x \le 3$ *without* performing any calculations. Then, find the average rate of change for $g(x) = x^2 - 4x + 5$ on the interval $[-1, 2]$.

Problem 3. Suppose that $h(x) = \sqrt{x-9} + 5$. Find at least two possible combinations for $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$. Find the domain and range for $h(x)$.

Problem 4. Let $f(x)$ be the function given by $f(x) = 3x - 7$.

(a) Find a value in the range of f. Be sure to justify why the value is in the range.

(b) Compute $f(4)$. Is $(4, 1)$ on the graph of f? Explain.

(c) Is there an x such that $f(x) = 11$? Explain.

(d) Is $1 \in f^{-1}(3)$? Explain.

(e) Assuming f^{-1} exists, what is $f(f^{-1}(\pi))$ and $f^{-1}(f(\sqrt{2}))$?